Prikshit Gautam

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GitHub – LinkedIn – Portfolio – Medium

Education

Thapar Institute of Engineering and Technology, Patiala
B.E. Electronics and Computer Engineering, CGPA: 8.56 — *Aug 2023 – Jun 2027*
St. Soldier Divine Public School, CBSE, Nangal
XII : **92.67%** — *May 2022*

Technical Skills

Languages: C++, Python, JavaScript, SQL
Libraries & Frameworks: PyTorch, Scikit-learn, TensorFlow, Keras, ONNX Runtime, LangChain, FAISS, Flask, NumPy, Pandas, Matplotlib, PyMuPDF
Developer Tools: AWS (EC2, S3), Render, Vercel, Git, GitHub (CI/CD), Postman, Streamlit, Hugging Face
AI Technology: NLP, RAG, LLM Integration, Vector Search, Time Series Forecasting, Model Optimization, Data Visualization
Databases: MySQL, DynamoDB
Relevant Coursework: Database Management Systems, DSA, Object Oriented Programming, Cloud Computing, OS, Computer Networks, Operating System

Experience

Undergraduate Research Assistant | TIET, Patiala — *Aug 2024 – May 2025*
Project: 5G IP Latency Predictor using Deep Sequential Methods | **LINK** | **LIVE**
Technology: PyTorch, Keras, Statistical Analysis

- Analyzed **40,000+** 5G packet traces from the **KTH Expeca testbed (Sweden)**, isolating RLC frame alignment limitations as a key structural source of network delay.
- Designed a hybrid LSTM model forecasting 5G latency with **94.68%** tracking accuracy (**R^2: 0.91**), mitigating multi-step error accumulation; co-authored a journal paper titled ***"Forecasting 5G Delay Using Deep Sequential Methods"*** under review.
- Reduced multi-step prediction variance by **45%** over baseline autoregressive variants using a custom sequential modeling loop configured for cellular edge environments.
- Optimized deployment via **tf2onnx**, dropping RAM from **450MB to 80MB**; deployed a Flask REST API for a model that predicts batch payloads of **3,470 packets in 2.5** seconds via an $O(N)$ execution loop.

Experiential Learning Centre | *Summer Project Intern* | **Credentials** — *Jun 2025 – Jul 2025*

- Led development of an automated animal-hazard visual recognition module using YOLOv8, OpenCV, and SUMO/TraCI to mitigate roadway traffic hazards.
- Coordinated system integration across traffic components, validating system logic matrices before a panel of 10+ faculty reviewers.

Projects

Healthcare Assistant Chatbot (MediQuery) | **GitHub** | **LIVE** — *Jan 2025 – Feb 2025*
Technology: Python, LangChain, FAISS, Groq LPU, Streamlit

- Built a healthcare RAG pipeline parsing a **759-page** encyclopedia into **7,470** chunks using **PyMuPDF** and **RecursiveCharacterTextSplitter**.
- Embedded text via **all-MiniLM-L6-v2** with **L2 normalization** for **40ms** CPU vector retrieval across **3.1M** characters.
- Streamed context passages to **Llama-3.1-8B-Instant** via **Groq LPUs** (temp **0.2**), delivering fact-grounded medical explanations in under **1.2 seconds**.
- Selected for an elite enterprise accelerator co-sponsored by **Microsoft & SAP** via Edunet Foundation.

AI-RoadIntelligence and Management System | **LINK** — *Jun 2025 – Jul 2025*
Technology: YOLOv8, OpenCV, SUMO, SightEngine API

- Narrowed a YOLOv8 80-class detector to **2** safety-critical classes (cow, horse) to isolate large road-blocking hazards.
- Constructed a frame-sampling pipeline running inference every **5th frame**, cutting compute overhead by **80%** with automated CSV logging.
- Integrated detections with a **SUMO/TraCI** controller, applying a **20s** presence threshold for alerts and a **60s** green-phase extension.

Achievements & Certifications

- **Adobe Hackathon — Top 3.5% Globally (Ranked 4,029/115,000 Teams)** [Certificate]
- **Complete Data Science and Machine Learning(under instructor Krish Naik (Udemy))** [Certificate]
- **Microsoft Azure AI Fundamentals: Generative AI** [Certificate]
- **Data Structures & Algorithms Proficiency** — *Ongoing*
  Solved **400+ DSA problems** to optimize runtime complexity across competitive platforms like **LeetCode**, **NeetCode** and **GeeksForGeeks**.